José Anastácio da Cunha — Principios Mathematicos (1790)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 17 October 2026.

Disclosure: prepared with AI assistance — the six subject reviewers named in the index are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To José Anastácio da Cunha — artillery lieutenant, professor at Coimbra by Pombal's reform, prisoner of the Inquisition, and then teacher of poor boys at the Real Casa Pia, for whose pupils the *Principios* were composed — under his own name, as everything in this series is published under the dead scholar's name. The folder counts his wait for an Englishing at 236 years (1790–2026), subject to the finding below.

The author's own terms

Desprezar sem erro notavel — “to disregard without appreciable error,” his hinge phrase in the page 106 definition of a convergent series. *Grandeza* is “magnitude,” any quantity, geometric or numeric; *contrarias*, “contrary” magnitudes — our positive and negative. *Potencia*, *expoente*, *raiz*, *base*, and *logarithmo* are power, exponent, root, base, and logarithm — *logarithmo* being defined by Cunha himself (page 119) as the exponent itself. His framing: “Mathematicians call a series convergent when…” — received usage, sharpened, not a claim of sole invention.

The story

José Anastácio da Cunha was born in Lisbon on 11 May 1744 and died there on 1 January 1787, aged forty-two. Educated by the Oratory in Lisbon and self-taught in mathematics and physics, he became a lieutenant of artillery in 1763, at nineteen, and served ten years. Under Pombal's reforms he was appointed professor of geometry at Coimbra in 1773. After King Joseph I died and Pombal fell, the Inquisition imprisoned him; in October 1778 he was sentenced to three years for following Voltaire and supporting heretical views. Freed in 1781, his health ruined, he became professor of mathematics at the College of São Lucas of the Real Casa Pia.

The work and what survives

The *Principios Mathematicos* were begun in parts from 1782 and published in full at Lisbon in 1790, after his death: twenty-one books of definitions, propositions, and proofs. The folder's collation gives 302 pages of text, and an errata list. The book survives in Portuguese. In this treatise Cunha gave his notion of a convergent series — one in which the remainder beyond any given term can be made arbitrarily small — and then defined exponentiation and logarithm through infinite series.

This project holds one book of the twenty-one. Its headings read Book VIII; Cunha's citations number it Book IX, and two successive propositions are both numbered VII (pages 114 and 117), uncorrected in the errata. The folder's text is a complete-coverage rendering of pages 100–120 — every page of the treatise is represented — in which long worked examples and parts of proofs are compressed into marked editorial summaries; it is not a sentence-by-sentence complete translation. It was made fresh from the 1790 Portuguese (Göttingen scan, below), never from the French, errata applied, formulas re-set in modern notation, and one ambiguous scan line (page 108) flagged. João Manuel d'Abreu's French translation (1811; Paris, 1816) is expressly not used: scholarship finds it inaccurate at that definition. The folder reports the dispute over it — Gonçalves (1940) claimed Cunha anticipated Cauchy; Youschkevitch (1973) contested it, on the faulty French text; Franco de Oliveira (1988), returning to the Portuguese, proposed the potential-infinity reading, concluding that Cunha's originality is thereby redressed — and takes no side beyond

printing that definition in English.

Its English fortunes

No English translation of any part of the *Principios* was found after research.

That sentence stands at that strength and no stronger. No WorldCat or union-catalogue sweep was run, and the English-language scholarship on the definition — Franco de Oliveira (1988) and Queiró (1988) — was not page-checked for quoted passages. If either renders the page 106 definition in quoted English, the claim narrows at passage level, without touching the standalone-translation finding.

What completing it would take

The folder refers at four load-bearing points to a NOTES.md absent from the folder — the errata log, the fact-check ledger, and the map of the untranslated remainder all live there by those references — so producing it, or retiring the references, is open work, and nothing parked there is verified. The rendering must still pass the folder's review gate (a historian of analysis; a reader of eighteenth-century Portuguese), undergo the passage-level check named above before any stronger English wording is used, and be collated against the Göttingen page images: page 108's ambiguous line, the errata-corrected definition of a power (page 109), and the duplicated Proposition VII numbering. The other twenty books remain untranslated.

Sources

- Project folder: `~/workspace/rare-and-obscure-books/books/cunha-principios-book9/cunha-principios-book9.txt` (base scan: SUB Göttingen, PPN590888331)
- PIN: `~/workspace/packages/boneyard-extension/research/PIN-cunha.md`
- MacTutor History of Mathematics, Anastácio da Cunha: https://mathshistory.st-andrews.ac.uk/Biographies/Cunha/